Mohammad Sharif

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See my website: https://msharif06.github.io/Portfolio

Education

Illinois Institute of Technology Chicago, IL

Bachelor of Science in Mechanical Engineering | GPA: 3.6 *August 2024 - December 2027*

- Relevant Coursework: Fluid Mechanics, Thermodynamics, Computational Mechanics, Advanced Mechanics of Solids, Material Science, Electromagnetism, Differential Equations, Multivariable Calculus
- Honors: Dean's List, Heald Scholar, STEM Scholar, Emerging Campus Leader Scholar

Experience

Project Engineering Co-Op, USG – Baltimore, MD *January 2026 - August 2026*

- Owned scope, planning, and execution of capital projects, including a $340,000 equipment upgrade and a $3,000,000 facility improvement initiative to enhance long-term operational reliability
- Coordinated contractors, project timelines, safety compliance, procurement, and communication with plant leadership throughout all project phases
- Spearheaded onboarding of engineering co-ops and completed a storage-focused CI project while supporting daily plant engineering operations

Biomedical Engineering Intern, Project C.U.R.E. – Woodridge, IL *August 2025 - December 2025*

- Ensured quality assurance of 100+ pieces of biomedical equipment from external medical facilities through inspection, testing, and repair using standard work instructions
- Contributed to 3 international medical shipments, including 60+ ICU beds, 15+ gurneys, 10+ home beds, infusion pumps, oxygen concentrators, and electrosurgical units to countries in need

Student Researcher, Illinois Institute of Technology – Chicago, IL *July 2025 - January 2026*

- Designed a cost-effective produce slicer with simplified parts, developing CAD models and technical drawings in SolidWorks.
- Field-tested and prototyped 8+ iterations to optimize blade types and performance across multiple metrics
- Organized technical reports in Overleaf summarizing research and design outcomes for patent submission

Projects

Bio-Inspired Frogbot October 2025 - December 2025

- Designed and assembled frog-inspired quadruped robot in Inventor with 4 servo motors controlling all gaits
- Achieved a performance of .4 body lengths per second for locomotion speed
- Optimized Arduino Uno code by determining servo max speeds and optimal torque application for efficient movement
- Created circuit integration connecting leg servos and batteries to Arduino controller

EV Charging Robotic Arm September 2025 - November 2025

- Designed 4-bar linkage mechanism to autonomously reach EV charging port at designated distance
- Developed MATLAB optimization code to determine optimal ground locations, bar lengths, and joint angles for linkage geometry
- Fabricated mechanism through laser cutting and 3D printing, integrating Arduino-controlled servo with IR receiver for signal-based actuation and retraction
- Produced technical documentation in Overleaf detailing design process, calculations, and performance analysis

iGEM Bee Husbandry Design Engineer & Treasurer August 2025 - December 2025

- Collaborated on the design of a bee monitoring enclosure for plasmid data collection
- Designed enclosure in SOLIDWORKS and supported fabrication via 3D printing, laser cutting, and CNC
- Served as project treasurer, managing budget tracking, and coordination with financials to report sending and ensure timely reimbursements

Skills

- Design & Simulation: SOLIDWORKS, AutoCAD, Inventor, DraftCAD, Orca Slicer
- Programming & Tools: Python, MATLAB, Arduino, LaTeX, Excel
- Languages: English, Urdu, Spanish